

Network layer protocols

- Communication protocols
 - Most of the IP-based communications are based on the IPV.4 (and often via gateway middleware solutions)
 - IP overhead makes it inefficient for embedded devices with low bit rate and constrained power.
 - However, IPv6.0 is increasingly being introduced for embedded devices
 - 6LowPAN will introduce significant optimization for network and transport functions on LP-WANs

HTTP, DNS, XMPP, MQTT, CoAp, TLS, SSL	Application layer
TCP, UDP	Transport Layer
IP(V4, V6), 6LowPAN	Internet Layer
Ethernet, 802.11 WiFi, 802.15.4	Link Layer
Electric, optical or RF transmission	Physical Layer

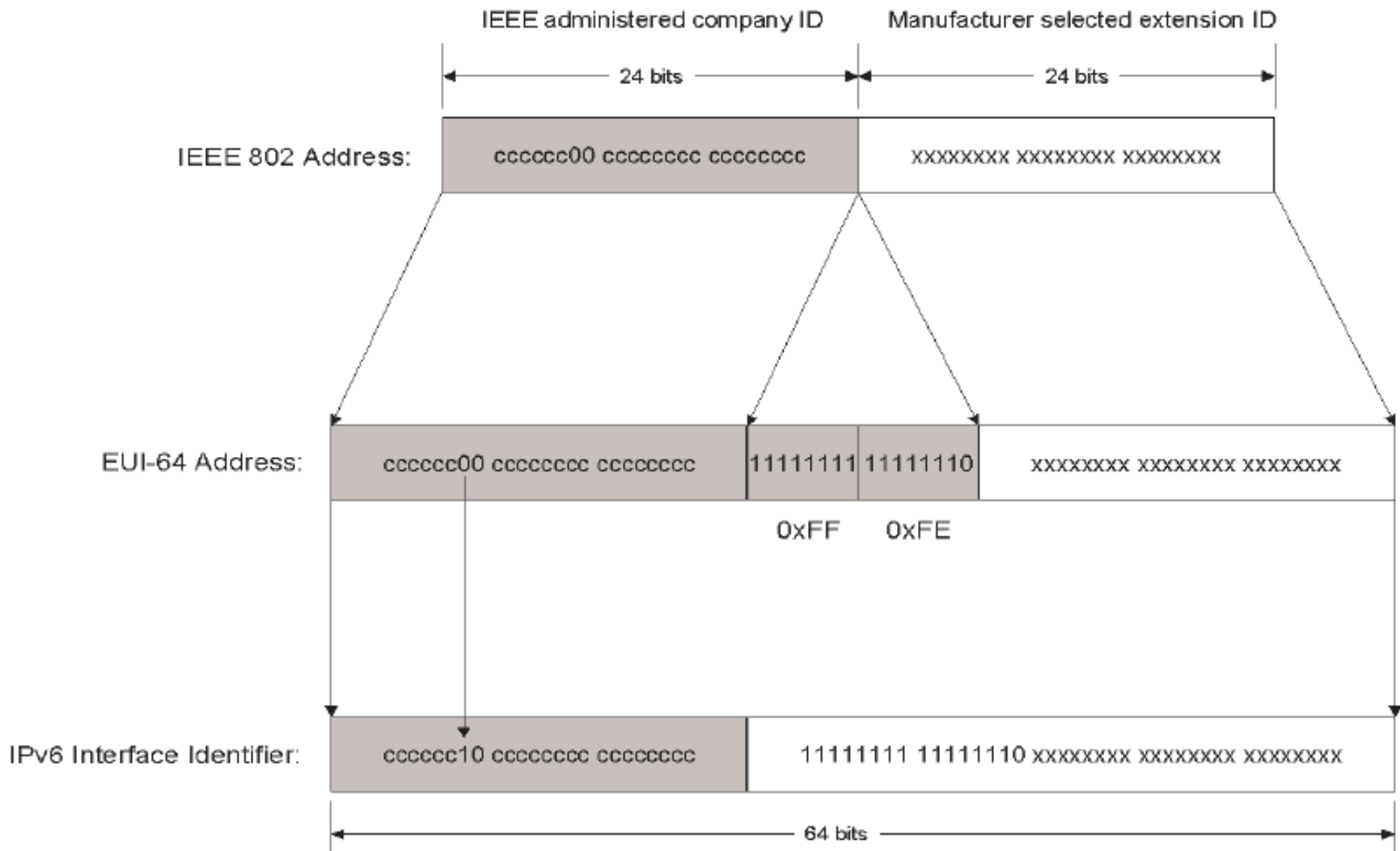
Internet Protocol v6

- IPv6 (RFC 2460) = the next generation Internet Protocol
 - Complete redesign of IP addressing
 - Hierarchical 128-bit address with decoupled host identifier
 - Stateless auto-configuration
 - Simple routing and address management
- Majority of traffic not yet IPv6 but...
 - Most PC operating systems already have IPv6
 - Governments are starting to require IPv6
 - Most routers already have IPv6 support
 - So the IPv6 transition is coming
 - 1400% annual growth in IPv6 traffic (2009)

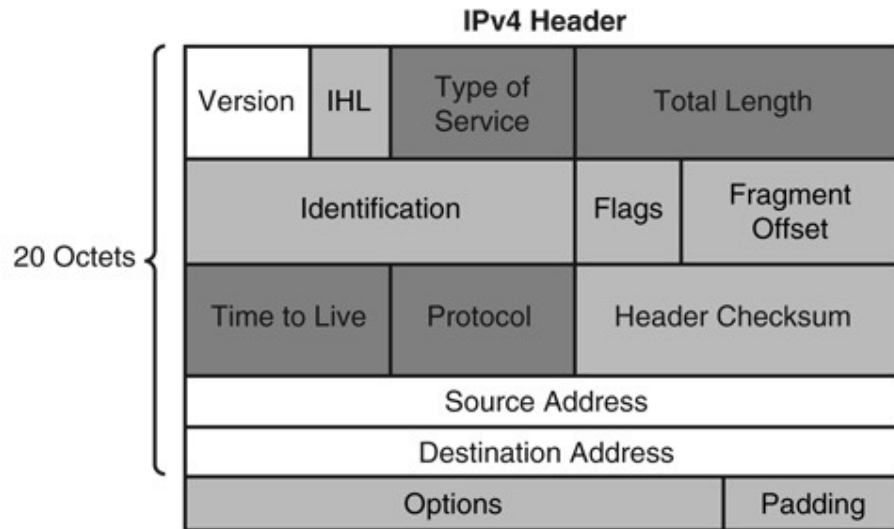
IPv6 Addressing

- 128-bit IPv6 address = 64-bit prefix + 64-bit Interface ID (IID)
- The 64-bit prefix is hierarchical
 - Identifies the network you are on and where it is globally
- The 64-bit IID identifies the network interface
 - Must be unique for that network
 - Typically is formed statelessly from the interface MAC address
 - Called Stateless Address Autoconfiguration (RFC2462)
- There are different kinds of IPv6 addresses
 - Loopback (0::1) and Unspecified (0::0)
 - Unicast with global (e.g. 2001::)
 - link-local (FE80::) for use during auto-configuration on p2p links
 - Site-local (FEC0::) Equivalent to the IPv4 private address space
 - Multicast addresses (starts with FF::)
 - Anycast addresses (special-purpose unicast address)

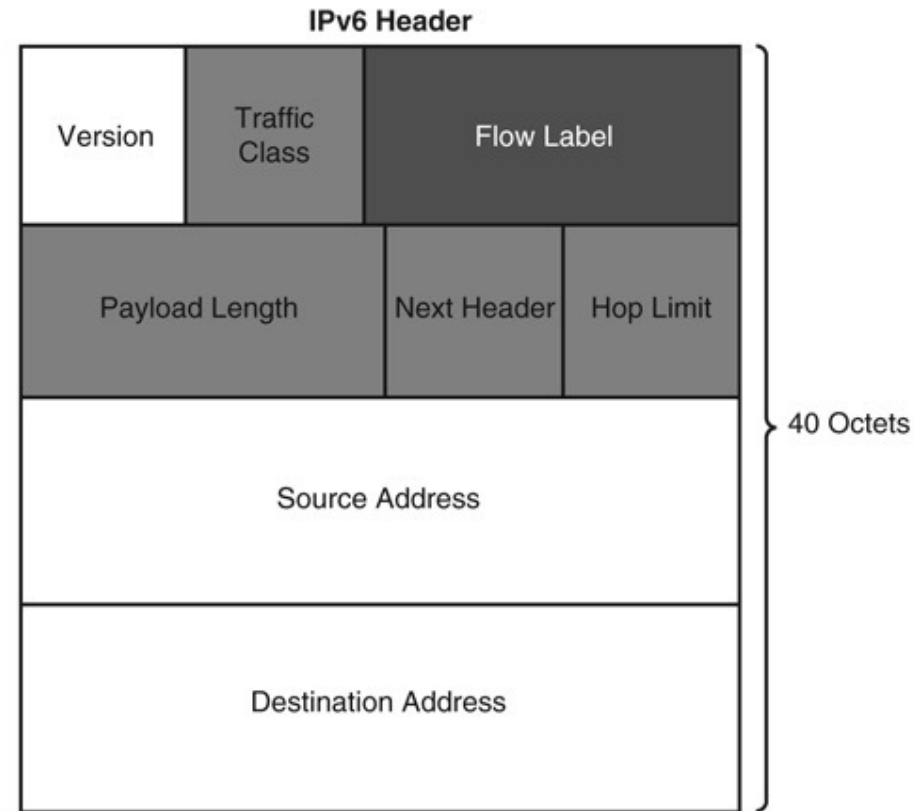
IEEE 802 → IPv6 Interface ID



IPv4 vs. IPv6 Header

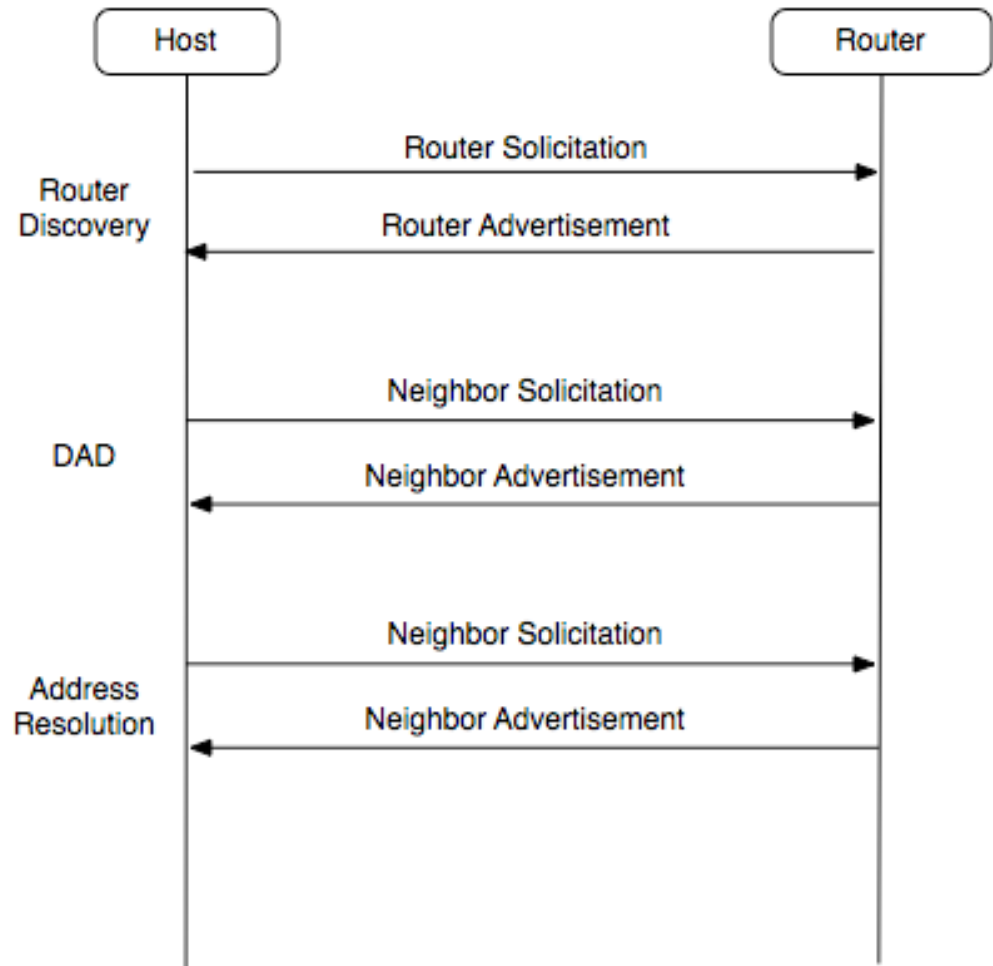


-  Fields name kept from IPv4 to IPv6.
-  Fields not kept in IPv6.
-  Field name and position changed in IPv6.
-  New field in IPv6.



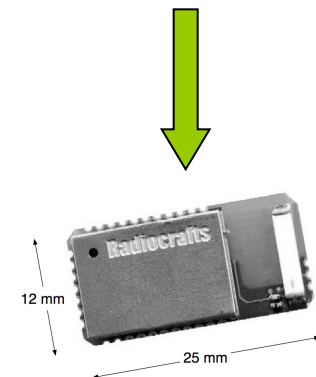
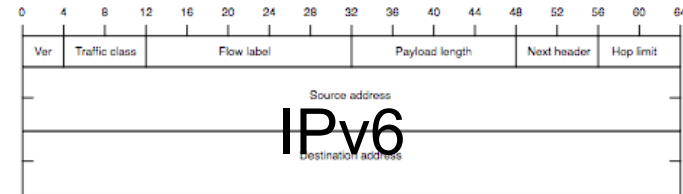
IPv6 Neighbor Discovery (ND)

- Neighbor Discovery (ND) is the basic enabler for IPv6 on IoT
 - “One-hop routing protocol” defined in RFC4861
- Defines the interface between neighbors
- Finding Routers
 - Router Solicitation (RS) / Router Advertisement (RA)
- Finding Neighbors
 - Neighbor Solicitation (NS) / Neighbor Acknowledgement (NA)
- Address resolution using NS/NA
- Detecting Duplicate Addresses using NS/NA
- Neighbor Unreachability Detection using NS/NA



IPv6 over Low power Wireless Personal Area Networks (6LowPAN)

- IPv6 requires the link to carry a payload of up to 64KB (or 1280 bytes without fragmentation)
- Low-power radio links do not support such a large payload
- **IEEE 802.15.4 frame only supports 127 Bytes of payload** and 81 Bytes in the worst case (with extended addressing and full security information)
- Defined by IETF standards
 - RFC 4919, 4944
 - draft-ietf-6lowpan-hc and -nd
 - draft-ietf-roll-rpl
- Direct end-to-end Internet integration
 - Multiple topology options



IPv6 over Low power Wireless Personal Area Networks (6LoWPAN)

- 6LoWPAN typically includes devices that work together to connect the physical environment to real-world applications, e.g., wireless sensors.
- Ensures Minimal use of code and memory
- Useful with low-power link layers such as IEEE 802.15.4, narrowband ISM and power-line communications
- Low bandwidth
 - Data rates of 250 kbps, 40 kbps, and 20 kbps for each of the currently defined physical layers (2.4 GHz, 915 MHz, and 868 MHz, respectively).
- Small packet size
 - the maximum physical layer packet is 127 bytes
 - 81 octets ($81 * 8$ bits) for data packets.

IPv6 over Low power Wireless Personal Area Networks (6LoWPAN)

- Support for both 16-bit short or IEEE 64-bit extended media access control addresses.
- Header compression
 - IPv6 base and extension headers, UDP header
- Fragmentation and reassembly
 - 6LoWPAN defines a header encoding to support fragmentation when IPv6 datagrams do not fit within a single frame and compresses IPv6 headers to reduce header overhead.
 - 1280 byte IPv6 MTU -> 127 byte 802.15.4 frames
- Network autoconfiguration using neighbor discovery
- Unicast, multicast and broadcast support
 - Multicast is compressed and mapped to broadcast
- Support for IP routing (e.g. IETF RPL)
- Support for use of link-layer mesh (e.g. 802.15.5)

The 6LoWPAN Protocol Stack

TCP/IP Protocol Stack

HTTP		RTP	
TCP	UDP	ICMP	
IP			
Ethernet MAC			
Ethernet PHY			

Application

Transport

Network

Data Link

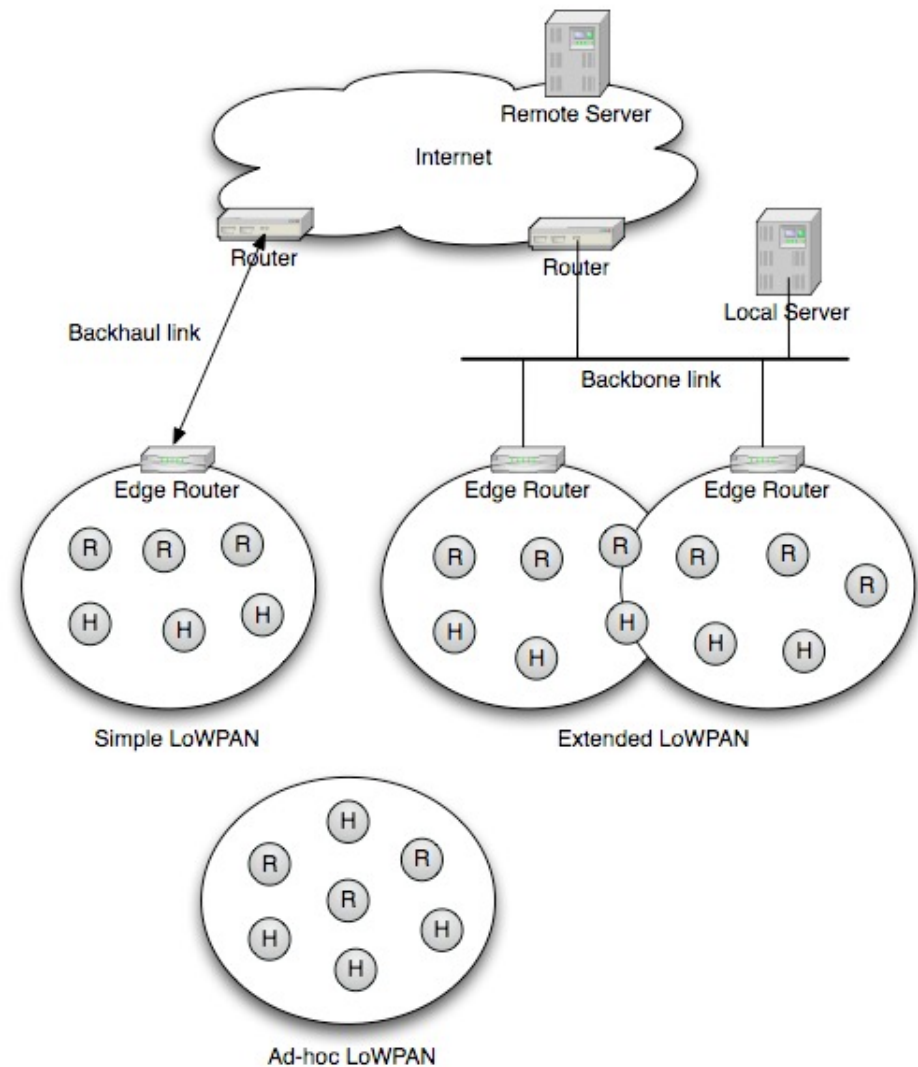
Physical

6LoWPAN Protocol Stack

Application	
UDP	ICMP
IPv6 with LoWPAN	
IEEE 802.15.4 MAC	
IEEE 802.15.4 PHY	

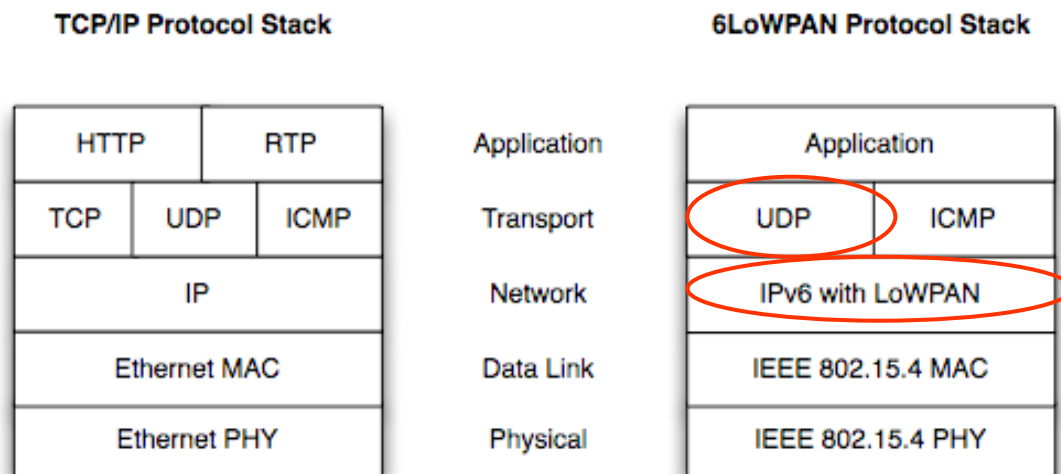
Architecture

- LoWPANs: stub networks (no transit) whose nodes share a common IPv6 prefix and register on an edge router
- Simple LoWPAN
 - Single Edge Router
- Extended LoWPAN
 - Multiple Edge Routers with common backbone link
- Ad-hoc LoWPAN
 - No route outside LoWPAN
- Internet Integration issues
 - Maximum transmission unit
 - Application protocols
 - IPv4 interconnectivity
 - Firewalls and NATs
 - Security



The 6LoWPAN Format

- 6LoWPAN is an adaptation header format
 - Enables the use of IPv6 over low-power wireless links
 - IPv6 header compression
 - UDP header compression
- Format initially defined in RFC4944
- Updated by draft-ietf-6lowpan-hc (work in progress)



Compression Principles 1/2

- 6LoWPAN compresses the header reducing redundancy
 - Some information is deducted from underlying link layer
- This achieves an efficient transport of IPv6 headers and next headers
 - In some cases, IPv6 addresses can be deduced from MAC addresses
 - IP payload length can be deduced from L2/L1 length information
 - Traffic Class and Flow Label values are set to zero
 - Version field value is IPv6
 - Hop limit can be set to predefined values

Compression Principles 2/2

- Example of header compression

IPv6 base header fields

Version	Elided; v6 only
Traffic Class	Set to 0
Flow Label	Set to 0
Payload	Deduced from link info
Next header	Maybe compressed
Hop limit	Set to a known value
Source	Deduced from MAC address or compressed or uncompressed carried inline
Destination	

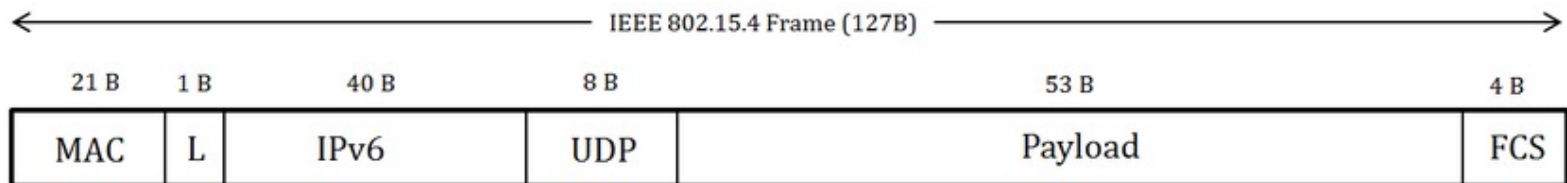
6LoWPAN

Potential IPv6 base header
fields to be carried inline

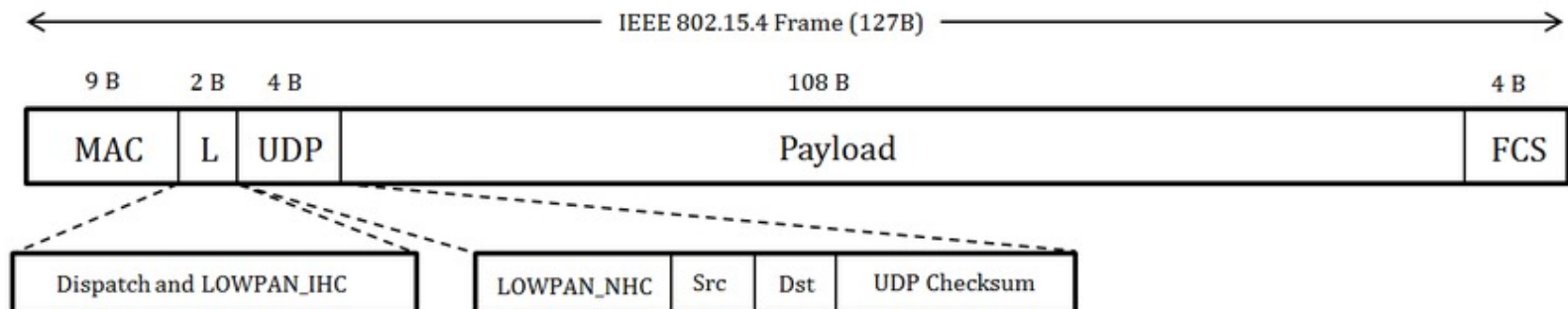
Version
Traffic class
Flow Label
Payload
Next header
Hop limit
Source
Destination

6LoWPAN Headers

- Orthogonal header format for efficiency
- Stateless header (L2+L3) compression from 70 to 15 bytes!
- Structure:
 - **Dispatch** value identifying the type of header,
 - IPv6 header compression byte (**LOWPAN_IHC**): which fields are compressed with in-line IPv6 fields.
 - UDP next-header compression byte (**LOWPAN_NHC**) with compressed source and destination port and UDP checksum (4 bytes)

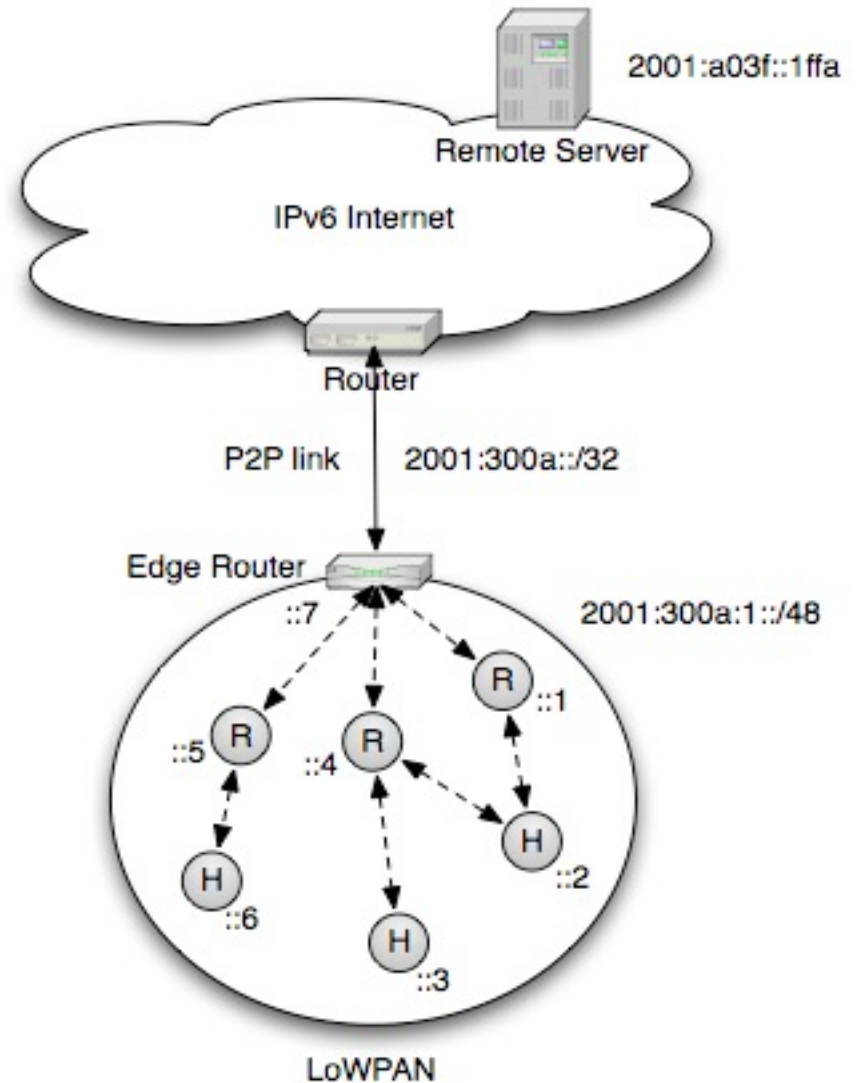


Full UDP/IPv6 (64-bit addressing)



6LoWPAN Addressing

- A LoWPAN works on the principle of
 - Nodes may behave as leaf hosts or intermediate routers in multi-hop relay arrangements
 - Only the edge router connects the LoWPAN to the IPv6 Internet
 - flat address spaces within lowpans (wireless network is one IPv6 subnet)
 - unique MAC addresses (e.g. 64-bit or 16-bit) within the subnet
 - Sharing a common prefix managed by the edge router

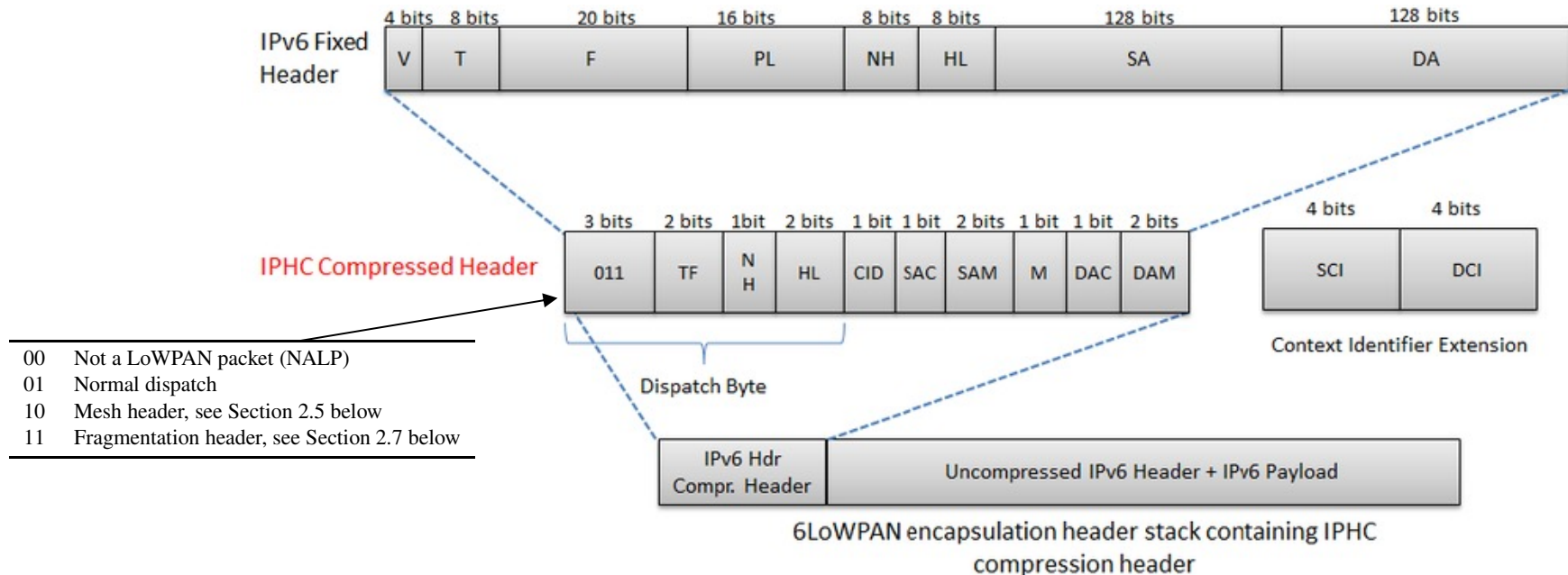


6LoWPAN Addressing

- 6LoWPAN suppresses or contracts all fields in the IPv6 header that can be simply derived from the 802.15.4 one:
 - Source address can be derived from datalink address
 - Destination can be address derived from datalink address
 - Length can be derived from link-layer frame length
 - Traffic class & flow label set to zero
- 6LoWPAN also compresses IPv6 addresses by
 - Eliding the IPv6 prefix
 - Global prefix known by all nodes in network
 - Link-local prefix indicated by header compression format
 - Compressing the Interface ID
 - Elided for link-local communication
 - Compressed for multihop dst/src addresses
 - Compressing with a well-known “context”
 - Multicast addresses are compressed

IP Header Compression (IPHC)

- IPv6 header and Next-header (UDP) compression
- Support for global address compression (with contexts)
 - CID flag controls whether a third byte is added to specify Context IDs for the source and destination address
- Support for IPv6 option header compression
- Support for compact multicast address compression



TF = Traffic Class, Flow Label

NH = Next Header Flag

HL = Hop Limit

CID = Context Identifier Extension flag

SAC = Source Address Compression

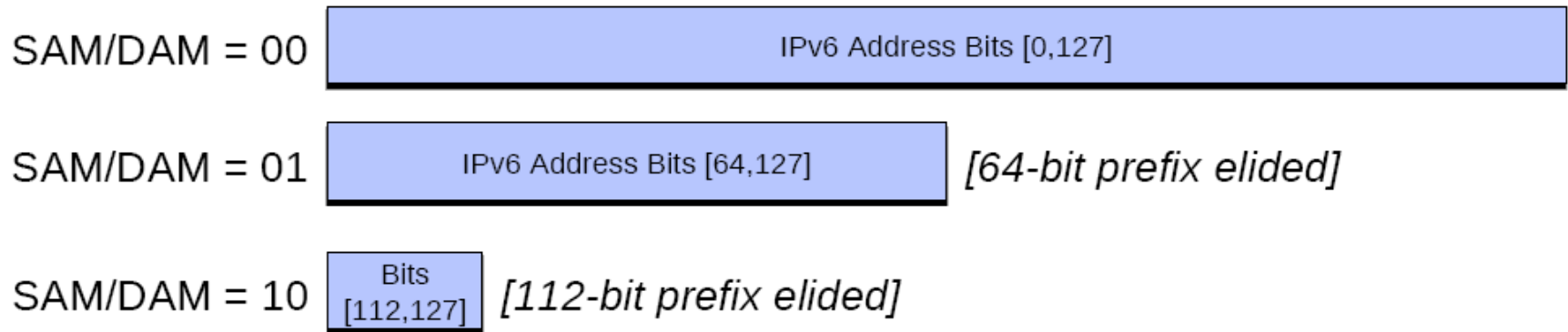
SAM = Source Address Mode

M = Multicast Compression

DAC = Destination Address Compression

DAM = Destination Address Mode

Address Compression



Stateless Mode (SAC/DAC=0)

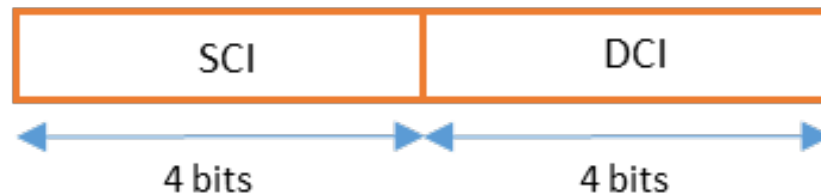
- Prefix is link-local (fe80::/10)

Context-Based (SAC/DAC=1)

- Prefix taken from stored contexts (up to 16 contexts)
- CID = 0, use ContextID = 0
- CID = 1, include 4-bit ContextID for source & destination

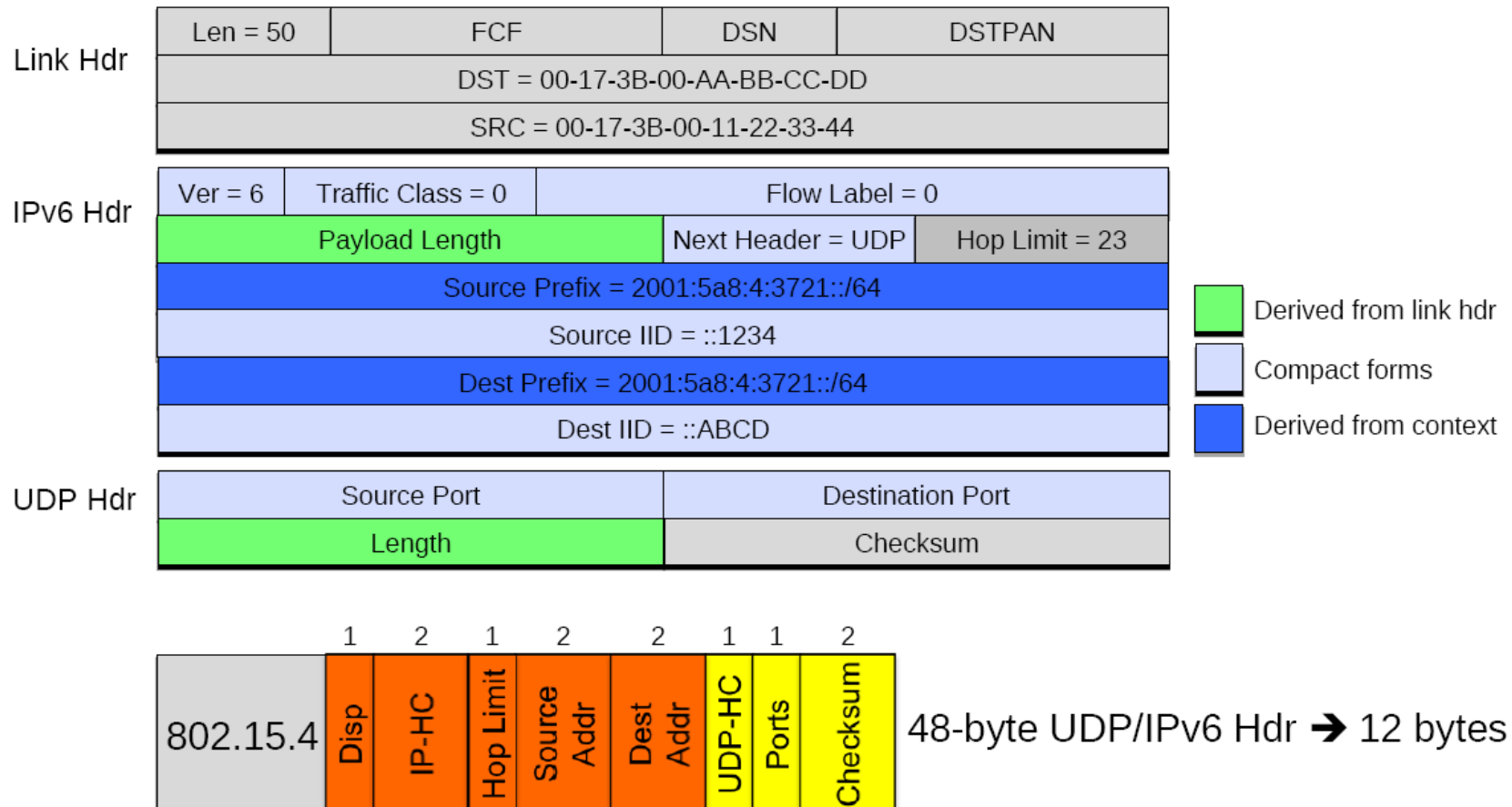
6LowPan Routing Context

- Contexts have a length and a context ID associated that must be synchronized in the entire Network.
- The first step in using context compression is to distribute the context to all the nodes in the Network by means of propagating the Network Data.

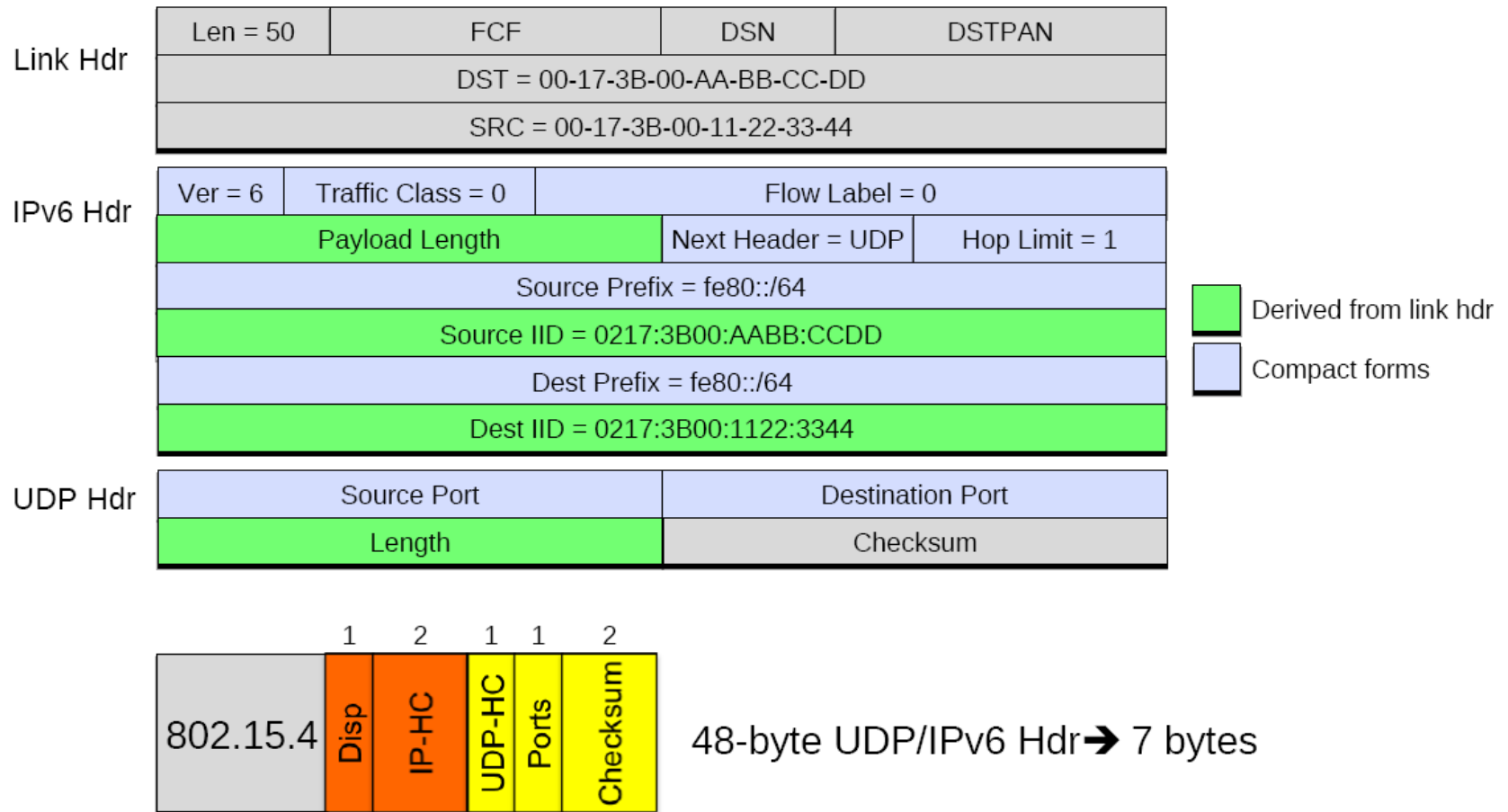


- Local Prefix
 - FD00:0DB8::/64 -> Context Id 0 (the first non-link local prefix in the network has ID 0)
- Global Prefixes
 - 2001::/64 -> Context Id 1
 - 2003::/64 -> Context Id 2

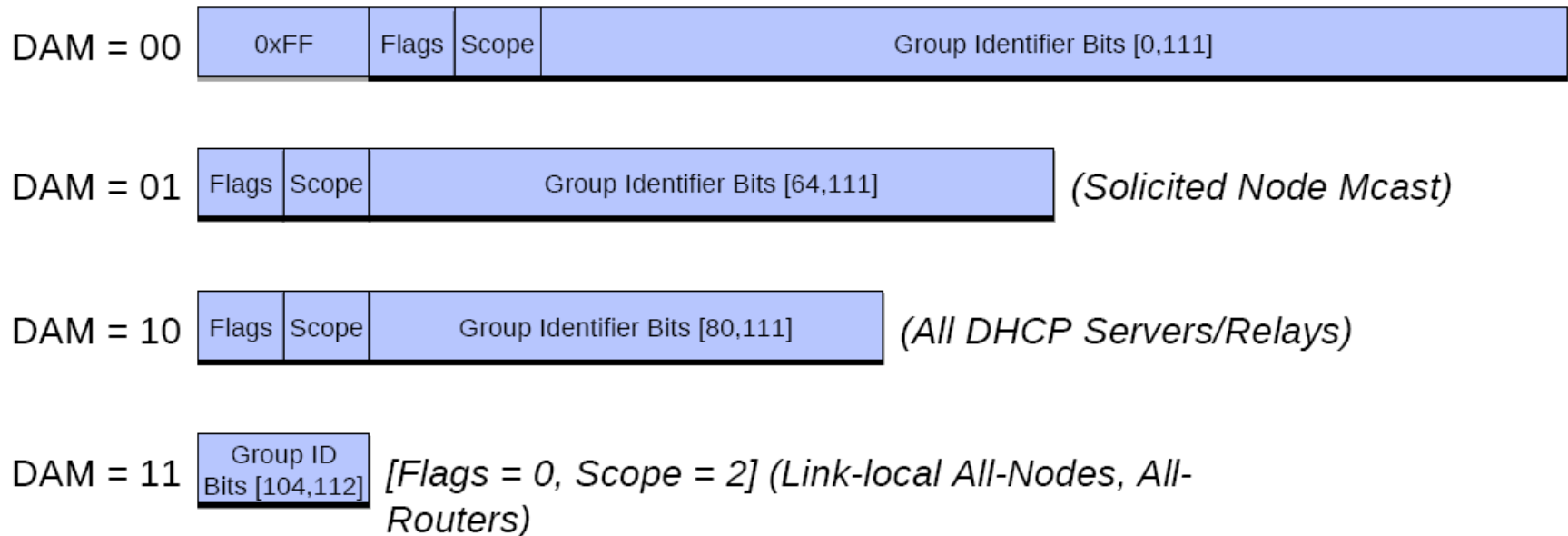
Unicast address mapping



Link local address mapping



Multicast Addresses Compression

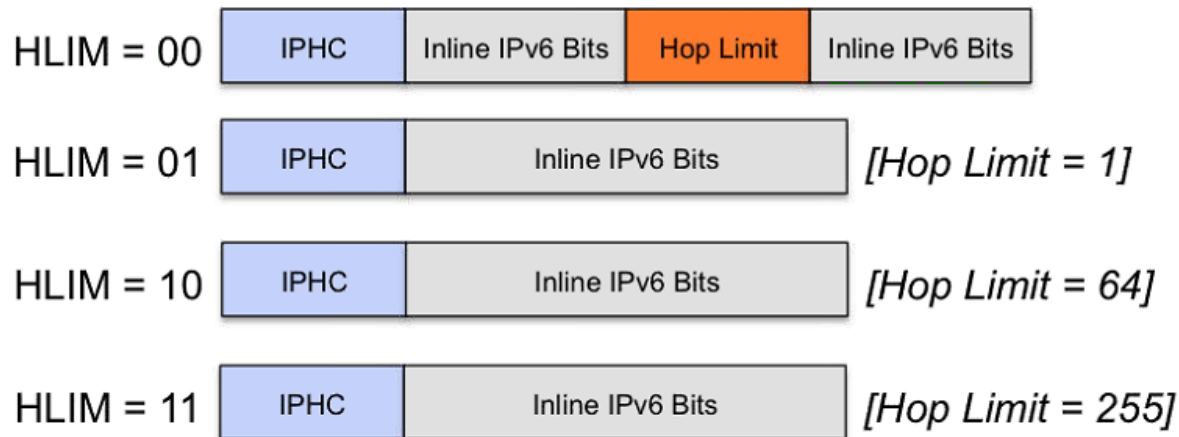


Next Header and Hop limit compression

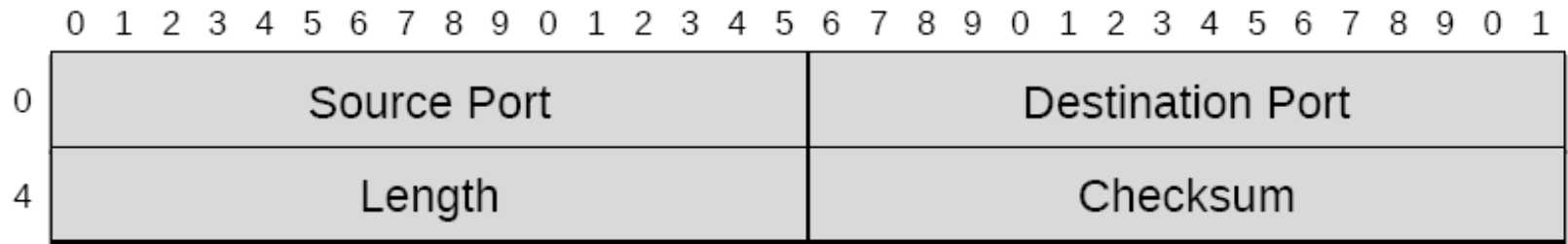
– Next Header Field



– Hop Limit Field

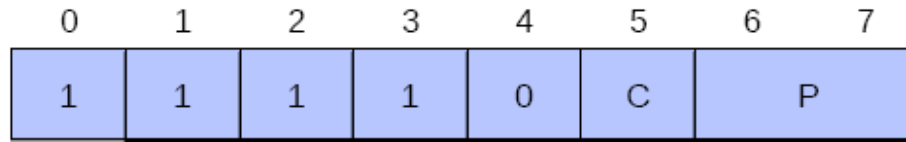


UDP Header Compression



Assume common values for header fields and define compact forms

- Ports within 61616 to 61632 (4 bits)
- Length derived from IPv6 Length



- Checksum may be elided if other integrity checks are in use
- C (Checksum): 0: Inline, 1: Elide
- P (Ports):
 - 0: Inline
 - 1: Elide first 8 bits of Dest Port
 - 2: Elide first 8 bits of Source Port
 - 3: Elide first 12 bits of Source and Dest Ports

6LoWPAN Setup & Operation

- Autoconfiguration is important in embedded networks
- In order for a 6LoWPAN network to start functioning three phases are necessary:
 - 1. Establish Link-layer connectivity between nodes (**commissioning**)
 - 2. Network layer address configuration, discovery of neighbors, registrations (**bootstrapping**)
 - 3. Routing algorithm sets up paths (**route initialization**)
 - 4. Continuous maintenance of 1-3

Link-layer Commissioning

- In order for nodes to communicate with each other, they need to have compatible physical and link-layer settings.
- Example IEEE 802.15.4 settings:
 - Channel, modulation, data-rate (Channels 11-26 at 2.4 GHz)
 - Usually a default channel is used, and channels are scanned to find a router for use by Neighbor Discovery
 - Addressing mode (64-bit or 16-bit)
 - Typically 64-bit is a default, and 16-bit used if address available
 - MAC mode (beaconless or super-frame)
 - Beaconless mode is easiest for commissioning (no settings needed)
 - Security (on or off, encryption key)
 - In order to perform secure commissioning a default key should already be installed in the nodes

6LoWPAN Neighbor Discovery

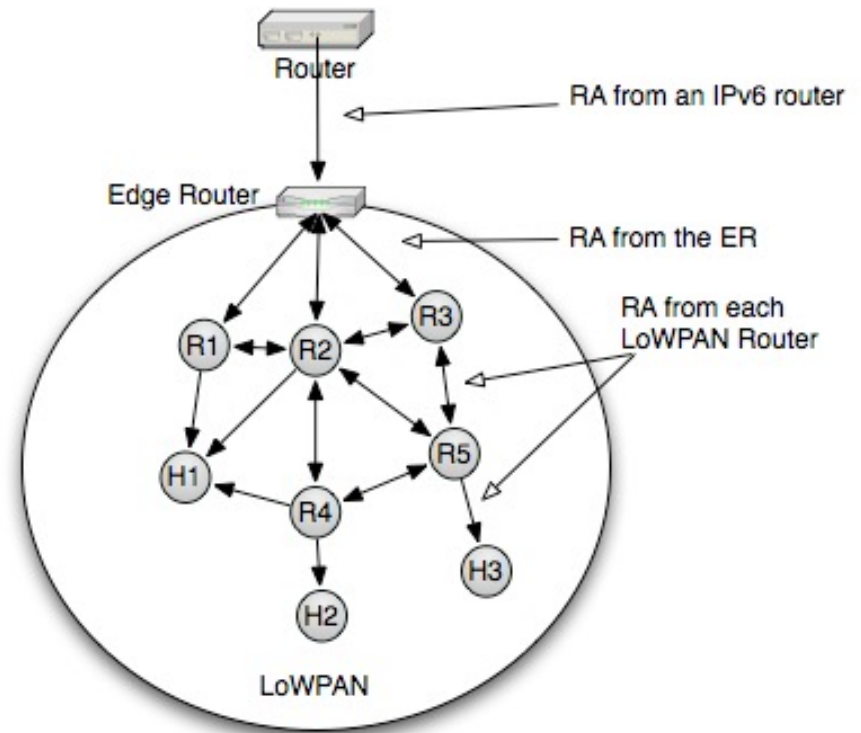
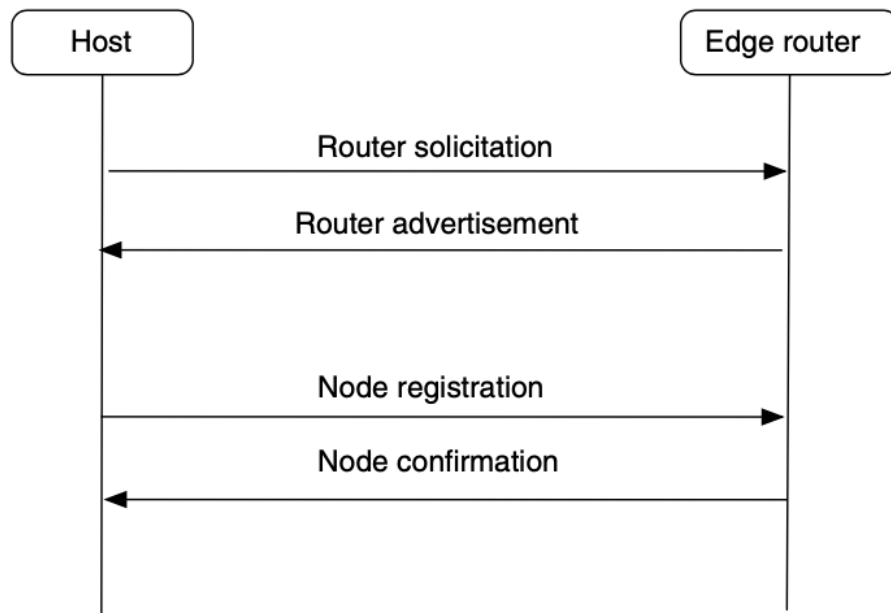
- Standard ND for IPv6 is not appropriate for 6LoWPAN:
 - Assumption of a single link for an IPv6 subnet prefix
 - Assumption that nodes are always on
 - Heavy use of multicast traffic (broadcast/flood in 6LoWPAN)
 - No efficient multihop support over e.g. 802.15.4
- The ND protocol divides nodes into host and router, where only router forwards packets that are not addressed to itself.
- Routers have to perform certain additional functions in ND as compared to hosts (route advertisements)
- 6LoWPAN Neighbor Discovery provides:
 - An appropriate link and subnet model for low-power wireless
 - Minimized node-initiated control traffic
 - Node Registration (NR) and Confirmation (NC)
 - Duplicate Address Detection (DAD) and recovery
 - Support for extended Edge Router infrastructures

Node Registration

- 6LoWPAN-ND Optimizes only the host-router interface
 - RFC4861 = signaling between all neighbors (distributed)
- Nodes register with their neighboring routers
 - Exchange of NR/NC messages
 - Binding table of registered nodes kept by the router
- Node registration exchange enables
 - Host/router unreachability detection
 - Address resolution (a priori)
 - Duplicate address detection
- Registrations are soft bindings
 - Periodically refreshed with a new NR message

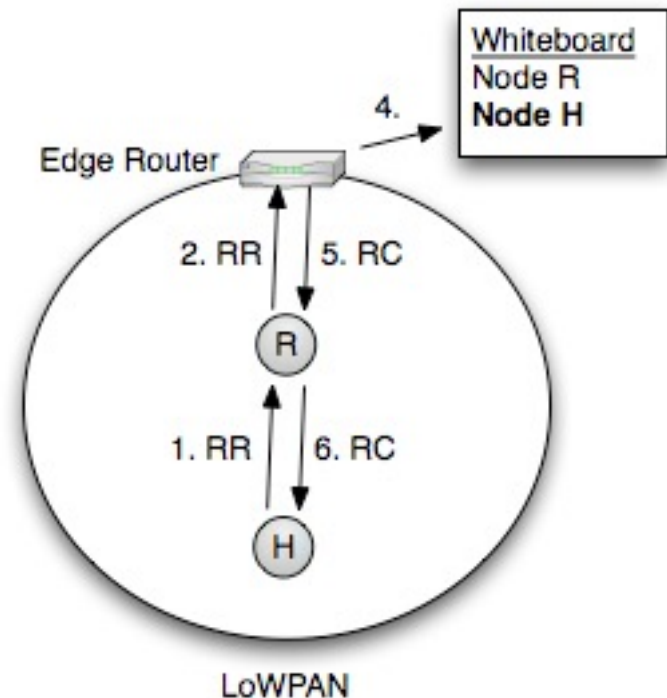
Prefix Dissemination

- In normal IPv6 networks RAs are sent to a link based on the information (prefix etc.) configured for that router interface
- In ND for 6LoWPAN RAs are also used to automatically disseminate router information across multiple hops



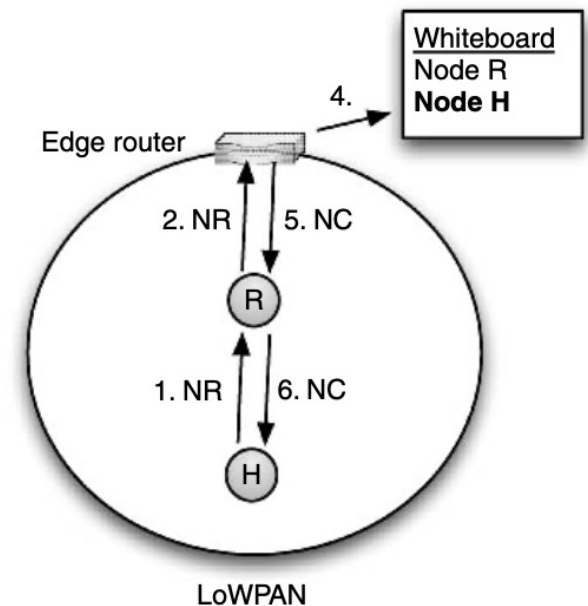
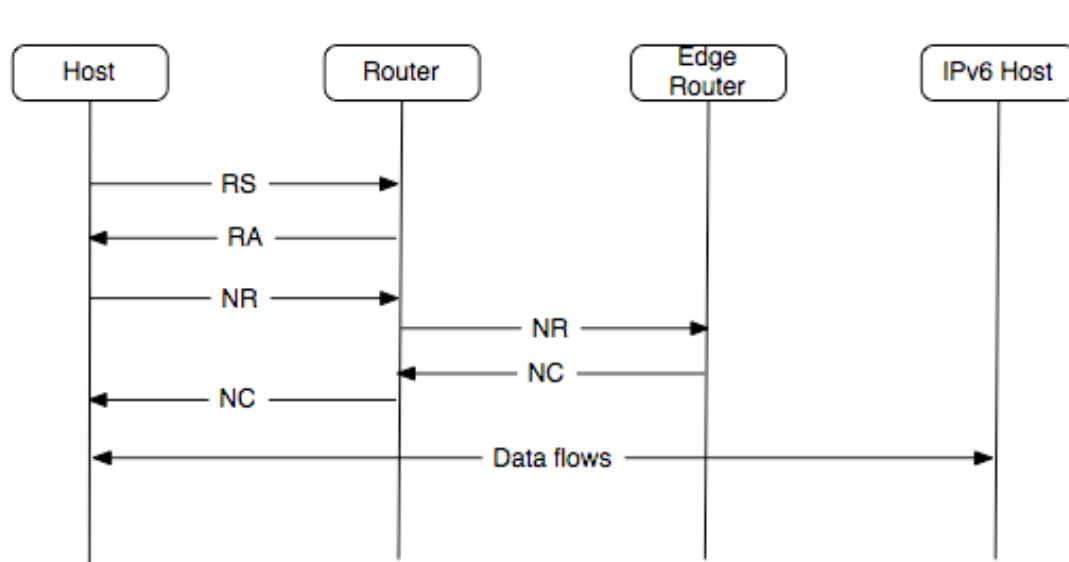
The 6lowpan whiteboard

- A whiteboard maintained by an edge router centralizes protocol state information
- The whiteboard is used in the LoWPAN for:
 - Duplicate address detection for the LoWPAN (= prefix)
 - Dealing with mobility (Extended LoWPANs)
 - Short address generation
 - Locating nodes



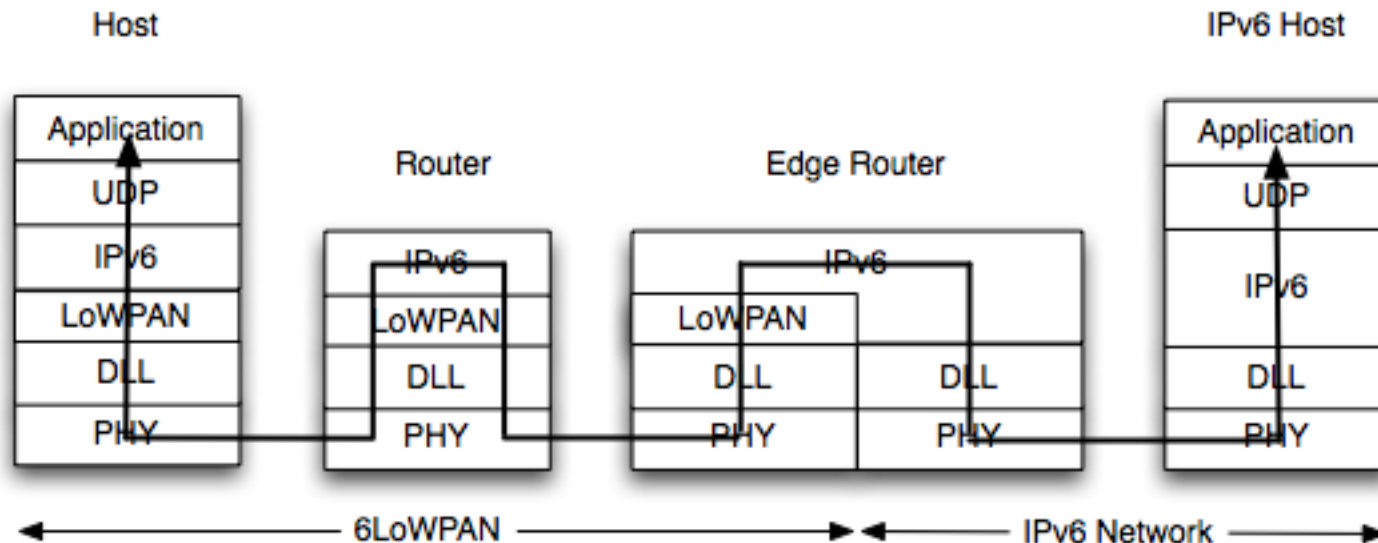
Multi-hop registration scenario

- The registration process becomes slightly more complicated when the node registering is not adjacent to an edge router



Routing in a Lowpan

- Single-interface routing
- Flat address space (exact-match)
- Stub network (no transit routing)



Extended LoWPANs

- Extended LoWPANs consist of two or more LoWPANs:
 - Which share the same IPv6 prefix
 - Which are connected together by a backbone link
- Whiteboards are synchronized over the backbone link

